

1 Summary

Model (Run-name): model3 (test)

Iteration: 11

TAC: 63897.480313900254

Took: 285.2812509536743s

Total Time: 933.6078970432281s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

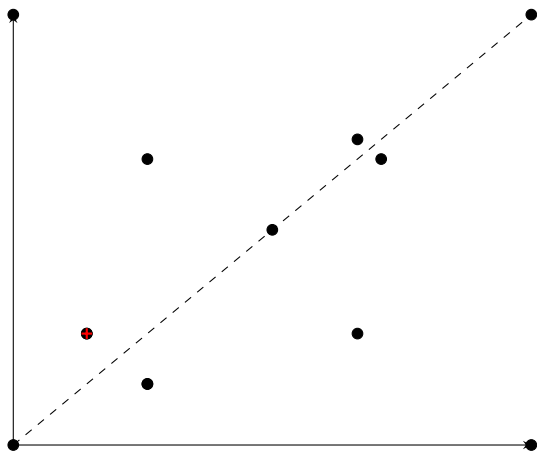
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{2,4,2}$	68.65838695604558	$q_{1,1,2}$	586.96
$z_{2,1,1}$	1.0	$dt_{2,4,3}$	55.6	$q_{2,1,1}$	173.039999999999496
$z_{2,2,1}$	1.0	$dt_{3,5,2}$	22.3	$q_{2,2,1}$	647.2100000000012
$z_{2,4,2}$	1.0	$dt_{3,5,3}$	29.05608108108041	$q_{2,4,2}$	357.4100000000003
$z_{3,5,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{3,5,2}$	1544.2900000000004
$z_{4,3,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{4,3,2}$	1539.72
$z_{5,4,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{5,4,1}$	1277.4400000000005
$z_{cu,3}$	1.0	$dt_{5,4,2}$	23.868711015316293	$q_{cu,3}$	839.98999999999867
$z_{cu,5}$	1.0	$dt_{cu,3}$	82.35608108108018	$q_{cu,5}$	1081.9699999999993
$A_{1,1,2}^\beta$	6.886731919061875	$dt_{cu,5}$	86.72824858757059	$t_{1,1}^{(H)}$	160.0
$A_{2,1,1}^\beta$	2.0956625614545397	$f_{1,1,2}^H$	8.8	$t_{1,2}^{(H)}$	160.0
$A_{2,2,1}^\beta$	5.790390532997608	$f_{2,1,1}^H$	1.5575157515751124	$t_{1,3}^{(H)}$	93.3
$A_{2,4,2}^\beta$	1.4876343666378133	$f_{2,2,1}^H$	5.825472547254831	$t_{2,1}^{(H)}$	248.9
$A_{3,5,2}^\beta$	12.782884632447086	$f_{2,4,2}^H$	7.14213592233036	$t_{2,2}^{(H)}$	171.51792452830125
$A_{4,3,2}^\beta$	6.675808728221597	$f_{3,5,2}^H$	9.585909373060211	$t_{2,3}^{(H)}$	137.8
$A_{5,4,1}^\beta$	12.091834947463191	$f_{4,3,2}^H$	12.6	$t_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	5.422947969797107	$f_{5,4,1}^H$	9.583195798949744	$t_{3,2}^{(H)}$	226.7
$A_{cu,5}^\beta$	6.203640738497231	$f_{1,1,2}^C$	5.869600000000002	$t_{3,3}^{(H)}$	122.35608108108018
$A_{1,1,2}$	24.928068483287923	$f_{2,1,1}^C$	7.6	$t_{4,1}^{(H)}$	271.1
$A_{2,1,1}$	3.4652942623256235	$f_{2,2,1}^C$	6.1	$t_{4,2}^{(H)}$	271.1
$A_{2,2,1}$	18.676744629514324	$f_{2,4,2}^C$	3.782116402116406	$t_{4,3}^{(H)}$	148.9
$A_{2,4,2}$	2.5223896397191172	$f_{3,5,2}^C$	13.9	$t_{5,1}^{(H)}$	198.9
$A_{3,5,2}$	71.16597394333384	$f_{4,3,2}^C$	8.4	$t_{5,2}^{(H)}$	126.72824858757059
$A_{4,3,2}$	23.66863193022334	$f_{5,4,1}^C$	17.3	$t_{5,3}^{(H)}$	126.72824858757059
$A_{5,4,1}$	65.27329951717095	$RecLMTD_{1,1,2}$	0.036099322289073754	$t_{1,1}^{(C)}$	160.0
$A_{cu,3}$	16.73899708924428	$RecLMTD_{2,1,1}$	0.01702207653130467	$t_{1,2}^{(C)}$	137.23157894736977
$A_{cu,5}$	20.944873037188774	$RecLMTD_{2,2,1}$	0.02452872009871173	$t_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	22.768421052631616	$RecLMTD_{2,4,2}$	0.03262135922330017	$t_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{3,5,2}$	0.03917080202023768	$t_{2,2}^{(C)}$	115.6
$dt_{2,1,1}$	88.9	$RecLMTD_{4,3,2}$	0.013066230964519417	$t_{2,3}^{(C)}$	115.6
$dt_{2,1,2}$	34.28634558093148	$RecLMTD_{5,4,1}$	0.04343241529120353	$t_{3,1}^{(C)}$	221.1
$dt_{2,2,1}$	27.2	$RecLMTD_{cu,3}$	0.01693847251259896	$t_{3,2}^{(C)}$	221.1
$dt_{2,2,2}$	55.91792452830322	$RecLMTD_{cu,5}$	0.016454376814154285	$t_{3,3}^{(C)}$	37.8

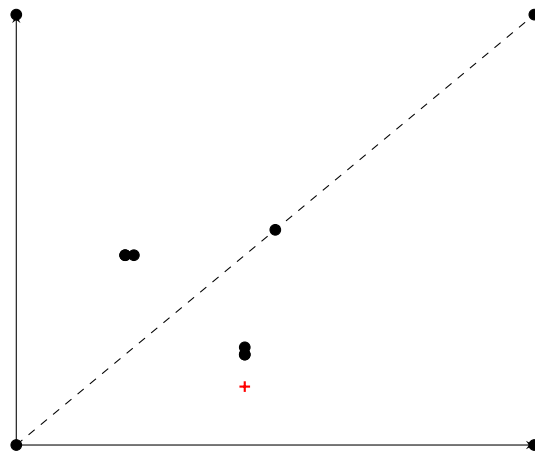
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,1}^{H,out}$	628.6576444111038
$t_{4,2}^{(C)}$	102.8595375722543	$b_{1,1,2}^{C,in}$	352.1760000000001
$t_{4,3}^{(C)}$	82.2	$b_{2,1,1}^{C,in}$	1042.9600000000114
$t_{5,1}^{(C)}$	204.4	$b_{2,2,1}^{C,in}$	705.1599999999879
$t_{5,2}^{(C)}$	204.4	$b_{2,4,2}^{C,in}$	310.8899682539686
$t_{5,3}^{(C)}$	93.3	$b_{3,5,2}^{C,in}$	1296.87
$t_{1,1,2}^H$	93.3	$b_{4,3,2}^{C,in}$	317.52
$t_{2,1,1}^H$	139.0711116115888	$b_{5,4,1}^{C,in}$	1779.4699999999973
$t_{2,2,1}^H$	139.85985526856254	$b_{1,1,2}^{C,out}$	939.1360000000002
$t_{2,4,2}^H$	137.8	$b_{2,1,1}^{C,out}$	1216.0
$t_{3,5,2}^H$	65.6	$b_{2,2,1}^{C,out}$	1352.37
$t_{4,3,2}^H$	148.9	$b_{2,4,2}^{C,out}$	668.2999682539689
$t_{5,4,1}^H$	65.6	$b_{3,5,2}^{C,out}$	2841.1600000000003
$t_{1,1,2}^C$	160.0	$b_{4,3,2}^{C,out}$	1857.24
$t_{2,1,1}^C$	160.0	$b_{5,4,1}^{C,out}$	3056.91
$t_{2,2,1}^C$	221.7		
$t_{2,4,2}^C$	126.81811071033897		
$t_{3,5,2}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,4,1}^C$	176.7		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,1,1}^{H,in}$	387.6656705670455		
$b_{2,2,1}^{H,in}$	1449.9601170117276		
$b_{2,4,2}^{H,in}$	1341.5963300971248		
$b_{3,5,2}^{H,in}$	2173.12565487275		
$b_{4,3,2}^{H,in}$	3415.8600000000024		
$b_{5,4,1}^{H,in}$	1906.0976444111043		
$b_{1,1,2}^{H,out}$	821.0400000000004		
$b_{2,1,1}^{H,out}$	214.6256705670505		
$b_{2,2,1}^{H,out}$	802.7501170117156		
$b_{2,4,2}^{H,out}$	984.1863300971246		
$b_{3,5,2}^{H,out}$	628.8356548727497		
$b_{4,3,2}^{H,out}$	1876.1400000000024		

3 *RecLMTD* Tangent Points

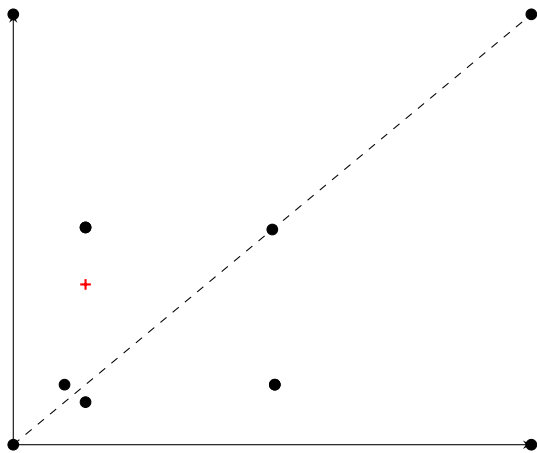
- (1, 1, 2)
 - Value: 0.036099
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000008
- (2, 1, 1)
 - Value: 0.017022
 - Error: 0.000423468282720
 - Relative error: 0.024273720725519
- (2, 2, 1)
 - Value: 0.024529
 - Error: 0.000565990635294
 - Relative error: 0.022554180492201
- (2, 4, 2)
 - Value: 0.032621
 - Error: 0.016466217397984
 - Relative error: -1.019255514809604
- (3, 5, 2)
 - Value: 0.039171
 - Error: 0.0000000000000001
 - Relative error: -0.0000000000000022
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000998024309
 - Relative error: 0.000076376124595
- (5, 4, 1)
 - Value: 0.043432
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000020
- $cu, 3$
 - Value: 0.016938
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000003
- $cu, 5$
 - Value: 0.016454
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



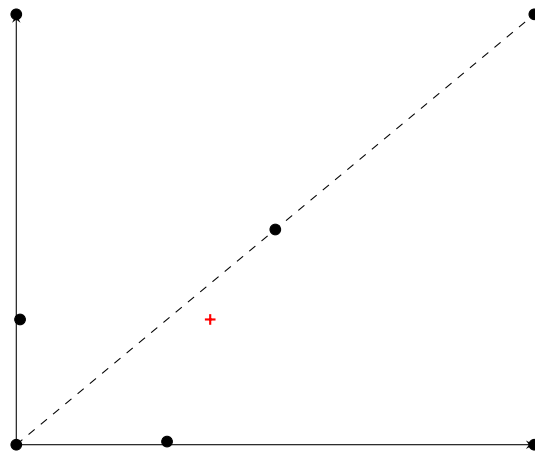
(a) RecLMTD - (1, 1, 2)



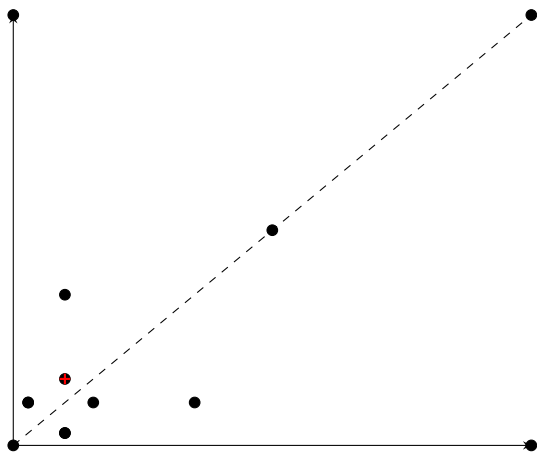
(b) RecLMTD - (2, 1, 1)



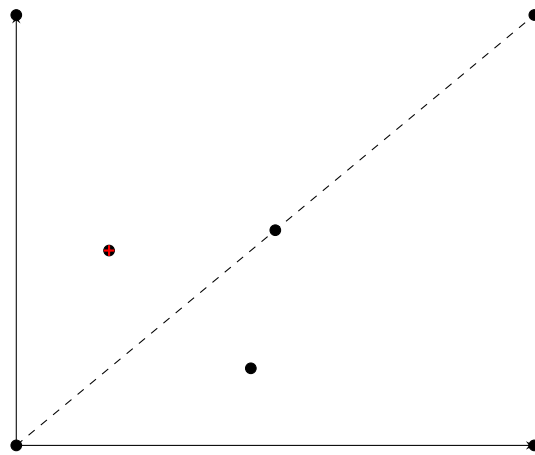
(c) RecLMTD - (2, 2, 1)



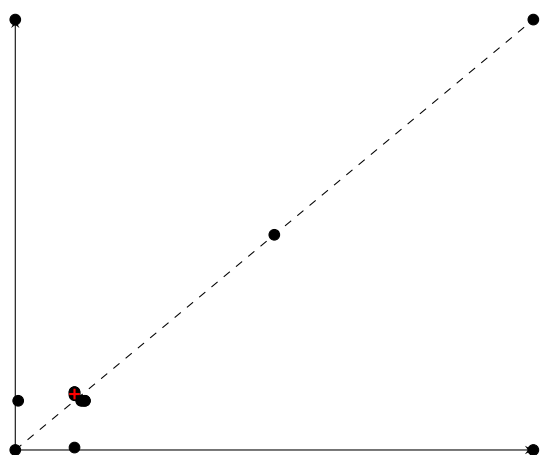
(d) RecLMTD - (2, 4, 2)



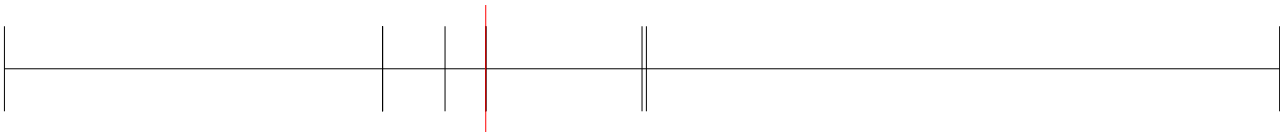
(e) RecLMTD - (3, 5, 2)



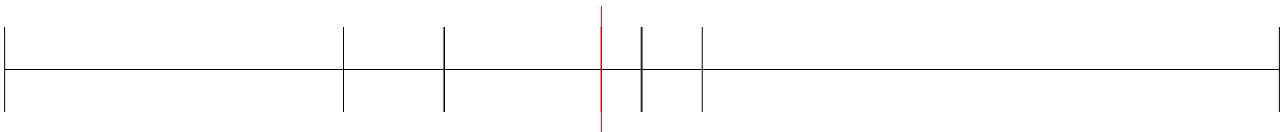
(f) RecLMTD - (4, 3, 2)



(g) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 387.6656705670455
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.9601170117276
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 1341.5963300971248
 - Absolute error: 116.59199999999578
 - Relative error: 0.09517680642871797
- (3, 5, 2)
 - Value: 2173.12565487275
 - Absolute error: 4.547473508864641e-13
 - Relative error: 2.092595749660378e-16
- (4, 3, 2)
 - Value: 3415.86000000000024
 - Absolute error: 2.2737367544323206e-12
 - Relative error: 6.656410843630361e-16
- (5, 4, 1)
 - Value: 1906.0976444111043
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.192875276405265e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.04000000000004
 - Absolute error: 3.410605131648481e-13
 - Relative error: 4.1540060553060515e-16
- (2, 1, 1)

- Value: 214.6256705670505
 - Absolute error: 1.9797763570595919
 - Relative error: -0.009140011874924026
- (2, 2, 1)
 - Value: 802.7501170117156
 - Absolute error: 11.999630318329423
 - Relative error: -0.014727995139185384
- (2, 4, 2)
 - Value: 984.1863300971246
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 628.8356548727497
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.807894270000029e-16
- (4, 3, 2)
 - Value: 1876.1400000000024
 - Absolute error: 2.2737367544323206e-12
 - Relative error: 1.2119227533298798e-15
- (5, 4, 1)
 - Value: 628.6576444111038
 - Absolute error: 6.821210263296962e-13
 - Relative error: 1.08504371559607e-15

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 352.1760000000001
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1042.9600000000114
 - Absolute error: 1.1368683772161603e-12
 - Relative error: 1.0900402481553933e-15
- (2, 2, 1)
 - Value: 705.1599999999879
 - Absolute error: 1.2050804798491299e-11
 - Relative error: -1.708946168031553e-14

- (2, 4, 2)
 - Value: 310.8899682539686
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 1779.4699999999973
 - Absolute error: 2.0463630789890885e-12
 - Relative error: -1.149984590349424e-15

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 939.1360000000002
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.2105471169416996e-16
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 668.2999682539689
 - Absolute error: 188.65911165098163
 - Relative error: 0.39333411458553097
- (3, 5, 2)
 - Value: 2841.1600000000003
 - Absolute error: 0.0

- Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



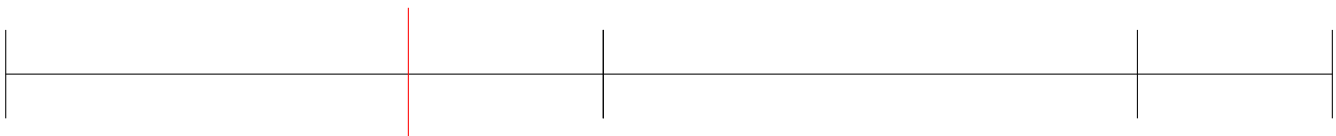
$$(a) \ t^{(H)} - (1, 2)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (2, 2)$$



$$(a) \ t^{(H)} - (3, 2)$$



$$(a) \ t^{(H)} - (4, 2)$$



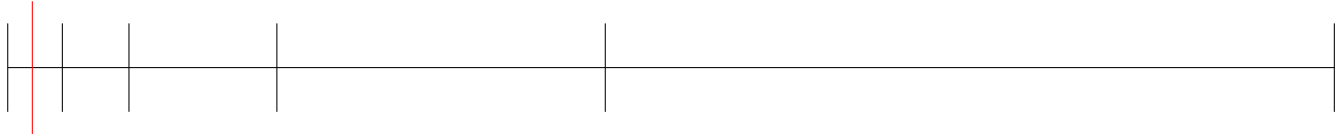
$$(a) \ t^{(H)} - (5, 1)$$



$$(a) \ t^{(H)} - (1, 1, 2)$$



(a) $t^{(H)} - (2, 1, 1)$



(a) $t^{(H)} - (2, 2, 1)$



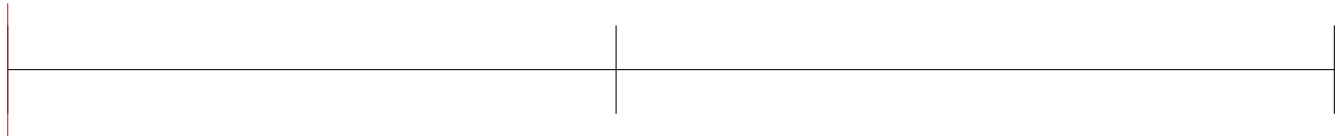
(a) $t^{(H)} - (2, 4, 2)$



(a) $t^{(H)} - (3, 5, 2)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 4, 1)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (1, 2)$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 2)$$



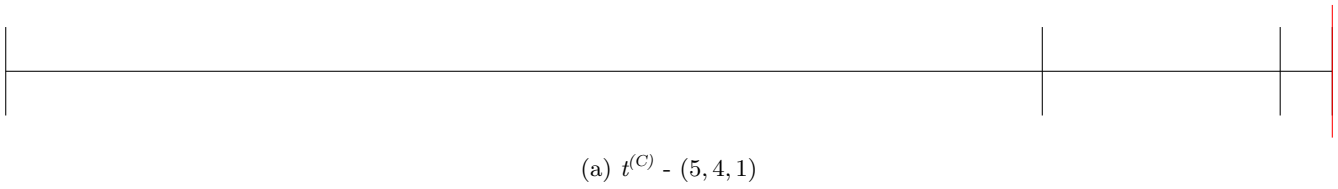
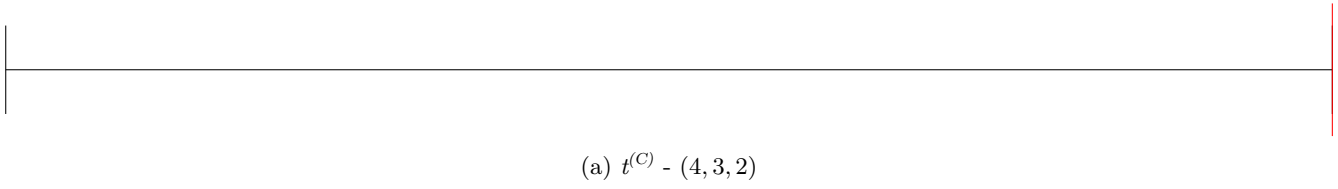
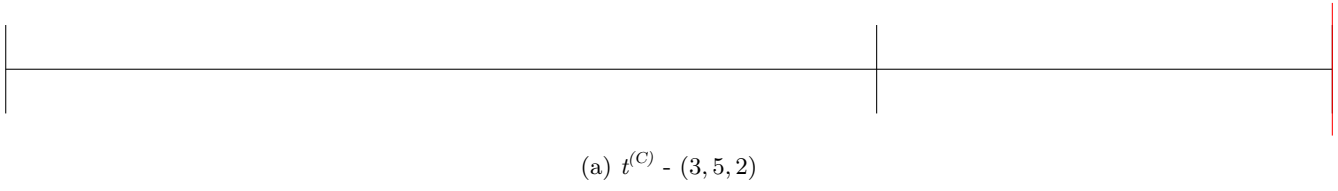
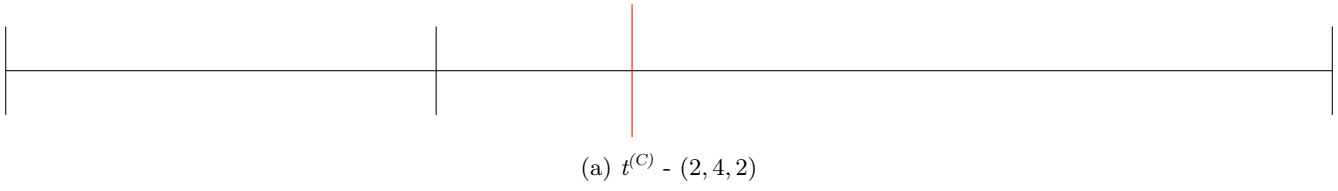
$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 1, 1)$$



$$(a) \ t^{(C)} - (2, 2, 1)$$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 24.928068
 - Error: 0.0000000000000004
 - Relative error: -0.0000000000000000
- $(2, 1, 1)$
 - Value: 3.465294
 - Error: 0.0000000000000111
 - Relative error: 0.0000000000000032
- $(2, 2, 1)$
 - Value: 18.676745
 - Error: 0.0000000000000398
 - Relative error: 0.0000000000000021
- $(2, 4, 2)$
 - Value: 2.522390
 - Error: 11.194316242633500
 - Relative error: 0.816108206930038
- $(3, 5, 2)$
 - Value: 71.165974
 - Error: 0.000000000001052
 - Relative error: -0.0000000000000015
- $(4, 3, 2)$
 - Value: 23.668632
 - Error: 0.0000000000000004
 - Relative error: -0.0000000000000000
- $(5, 4, 1)$
 - Value: 65.273300
 - Error: 0.0000000000000284
 - Relative error: -0.0000000000000004

5.2 A_{CU}

- $cu, 3$
 - Value: 16.738997
 - Error: 0.0000000000000163
 - Relative error: 0.0000000000000010
- $cu, 5$

- Value: 20.944873
- Error: 0.0000000000000050
- Relative error: 0.0000000000000002



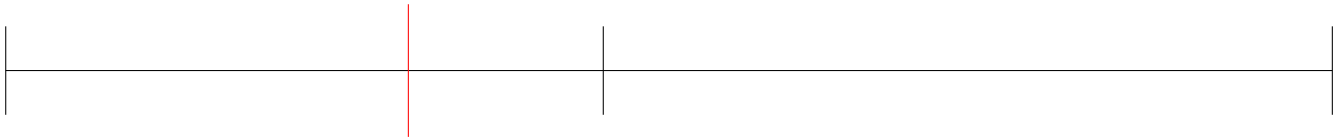
(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 4, 2)$



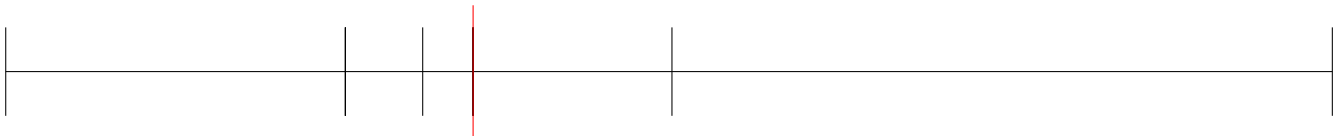
(a) $A^\beta - (3, 5, 2)$



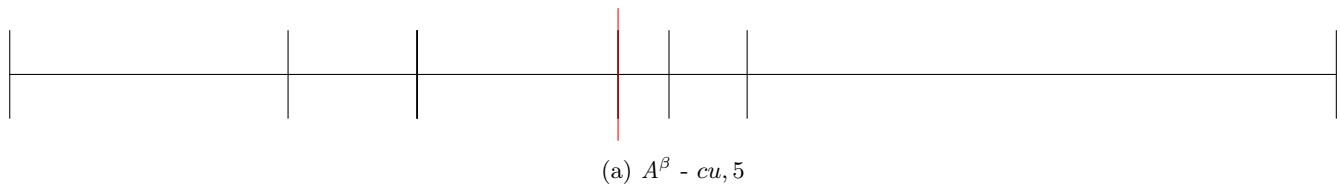
(a) $A^\beta - (4, 3, 2)$



(a) $A^\beta - (5, 4, 1)$



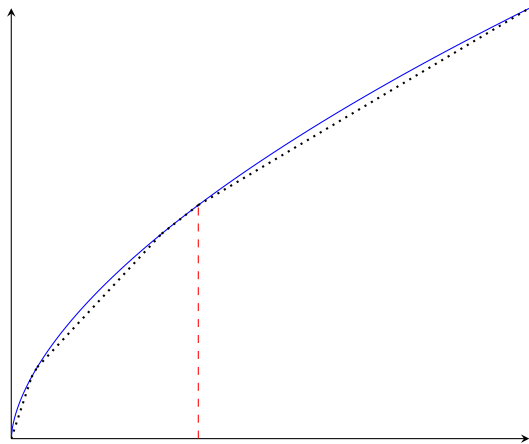
(a) $A^\beta - cu, 3$



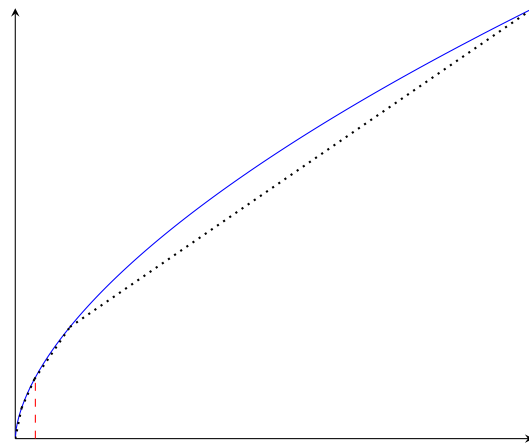
6 A^β Breakpoints

- (1, 1, 2)
 - q : 24.928068
 - $\hat{q}^\beta$: 6.886732
 - q^β : 6.886732
 - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.928068483288385, 69.05411764705883]
 - Error: 0.000000000000016
 - Relative error: -0.000000000000002
- (2, 1, 1)
 - q : 3.465294
 - $\hat{q}^\beta$: 2.095663
 - q^β : 2.107871
 - Breakpoints: [0, 1.0776943916793027, 3.3036464983028657, 9.439495705511714, 89.41176470588236]
 - Error: 0.012208646331359
 - Relative error: 0.005791931824992
- (2, 2, 1)
 - q : 18.676745
 - $\hat{q}^\beta$: 5.790391
 - q^β : 5.791362
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 18.890814020083003, 30.93278282491032, 30.932782824910646, 30.93278282491164, 76.14235294117645]
 - Error: 0.000971416412527
 - Relative error: 0.000167735399896
- (2, 4, 2)
 - q : 2.522390
 - $\hat{q}^\beta$: 1.487634
 - q^β : 1.742157
 - Breakpoints: [0, 3.743604220332404, 138.54823529411763]
 - Error: 0.254522689976818
 - Relative error: 0.146096294252257
- (3, 5, 2)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.782885
 - q^β : 12.923003
 - Breakpoints: [0, 8.941176470587976, 27.976182369080462, 55.82566076118323, 103.45829612760171, 181.68117647058827]
 - Error: 0.140118165879853
 - Relative error: 0.010842539312767

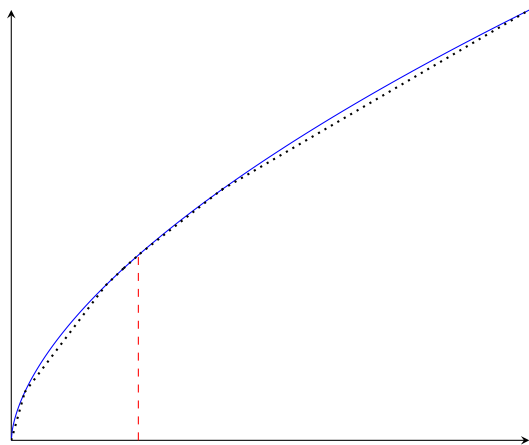
- (4, 3, 2)
 - q : 23.668632
 - $\hat{q}^\beta$: 6.675809
 - q^β : 6.675809
 - Breakpoints: [0, 19.92063386984238, 23.668654734065058, 23.668855770059224, 181.14352941176475]
 - Error: 0.000000132215718
 - Relative error: 0.000000019805198
- (5, 4, 1)
 - q : 65.273300
 - $\hat{q}^\beta$: 12.091835
 - q^β : 12.269909
 - Breakpoints: [0, 7.000222767756085, 12.878068451030797, 38.682379459353704, 82.70899711080247, 82.7196877037195, 192.3352941176471]
 - Error: 0.178073999532618
 - Relative error: 0.014513066095427
- $cu, 3$
 - q : 16.738997
 - $\hat{q}^\beta$: 5.422948
 - q^β : 5.422976
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.73892021814917, 16.738997089244634, 16.738997089244645, 18.852085704732907, 128.44338870924773]
 - Error: 0.000027754083642
 - Relative error: 0.000005117869792
- $cu, 5$
 - q : 20.944873
 - $\hat{q}^\beta$: 6.203641
 - q^β : 6.203641
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.235135805353295, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.0000000000000026
 - Relative error: 0.0000000000000004



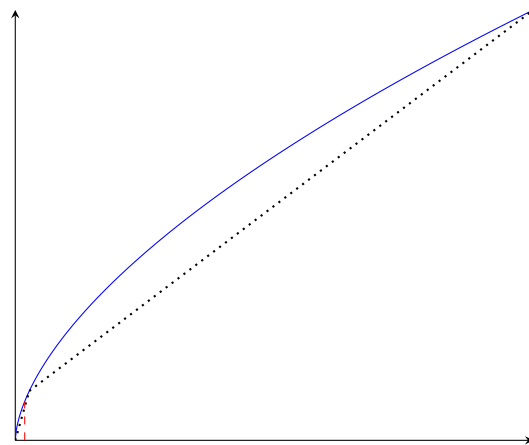
(a) $q^\beta - (1, 1, 2)$



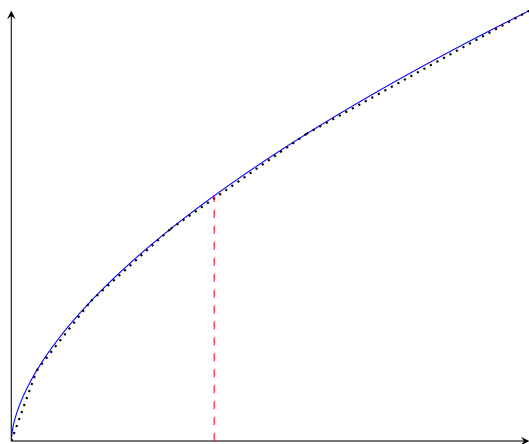
(b) $q^\beta - (2, 1, 1)$



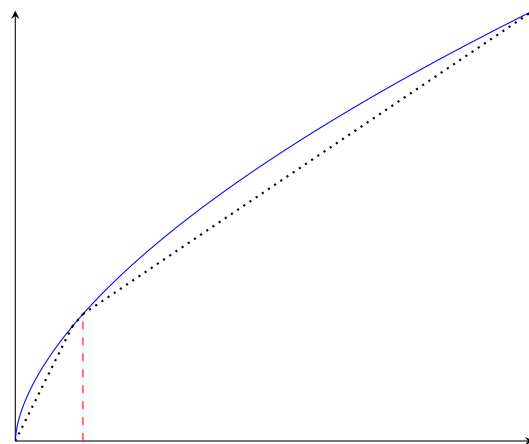
(c) $q^\beta - (2, 2, 1)$



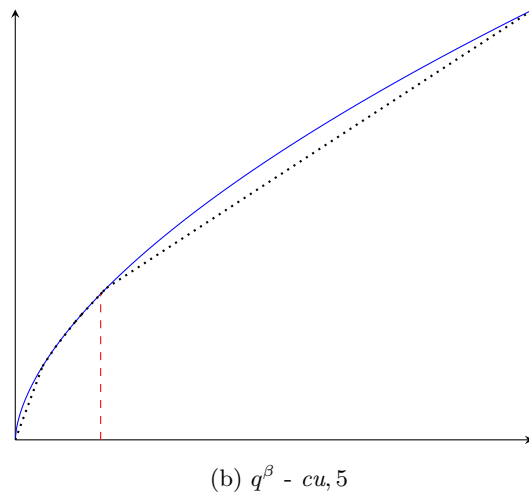
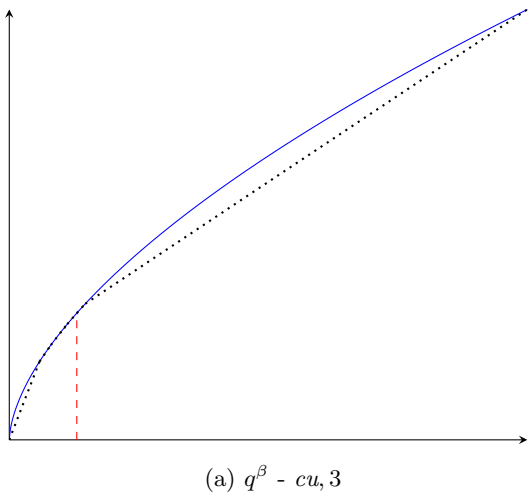
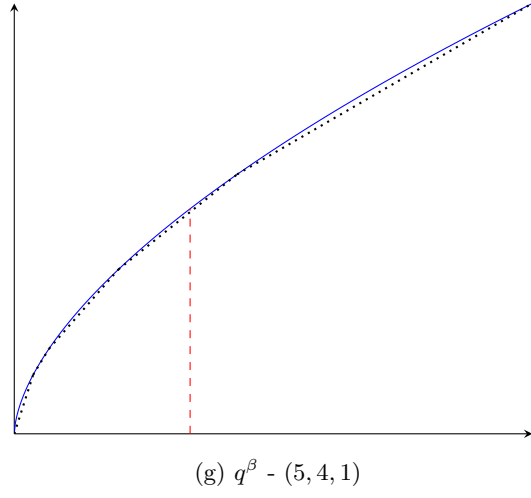
(d) $q^\beta - (2, 4, 2)$



(e) $q^\beta - (3, 5, 2)$



(f) $q^\beta - (4, 3, 2)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,2}$	-0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,2}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,2,2}$	-0.0	$z_{cu,1}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,2}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,1}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,3}$	77.8
$A_{hu,5}^\beta$	0.0	$A_{4,5,1}$	0.0	$dt_{2,2,3}$	22.2
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,3}$	100.0
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,5,2}$	10.0
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,3}$	44.5
$A_{1,5,1}$	0.0	$A_{5,4,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,1,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,2,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,1}$	0.0	$A_{cu,2}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,3,2}$	10.0
$A_{2,5,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,4,1}$	144.5
$A_{3,1,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,4,2}$	19.496543508826505
$A_{3,2,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	34.03716216216212
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,1}$	133.4
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{4,1,2}$	10.0
$A_{3,4,1}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,4,2}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,2,1}$	122.2
$A_{3,5,1}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	10.0
$A_{4,1,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	106.1
$A_{4,1,2}$	0.0	$dt_{1,4,2}$	77.8	$dt_{4,3,1}$	122.2
$A_{4,2,1}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,4,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,4,2}$	10.0
$A_{4,3,1}$	0.0	$dt_{1,5,2}$	10.0	$dt_{4,4,3}$	94.5

Variable	Value	Variable	Value	Variable	Value
$dt_{4,5,1}$	122.2	$f_{2,1,2}^H$	3.4578640776696337	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	10.0	$f_{2,2,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,1}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	10.0	$f_{2,5,1}^H$	3.217011701170056	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,2}^H$	0.0	$f_{1,5,1}^C$	13.9
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	14.8	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	5.214090626939731	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	0.0
$dt_{5,4,3}$	10.0	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	8.4
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,1}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	12.6	$f_{2,5,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	6.1
$dt_{hu,3}$	18.9	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	17.7	$f_{4,1,1}^C$	0.0
$f_{1,3,1}^H$	8.8	$f_{5,2,1}^H$	0.0	$f_{4,1,2}^C$	1.7303999999999996
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	8.116804201050257	$f_{4,2,2}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,3,1}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,2}^H$	0.0	$f_{4,4,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.1	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.024822011166820812	$q_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,5,1}$	0.1	$q_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.1	$q_{2,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,2,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.1	$q_{2,3,1}$	0.0
$f_{5,4,2}^C$	13.517883597883591	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,5,1}$	0.1	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02173396634811829	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.1	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.1	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.012853470437017995	$\text{RecLMTD}_{5,3,2}$	0.028842031741256373	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,2}$	0.06251525749564948	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.1	$q_{3,5,1}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,1}$	0.016108908592626606	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.07326370111710542	$q_{1,2,1}$	0.0	$q_{5,1,1}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0	$q_{5,1,2}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,1}$	0.0	$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	220.10241115115895	$t_{2,1,2}^C$	60.0
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	131.27996767618117	$t_{2,3,1}^C$	221.1
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	163.6020270270323	$t_{2,4,1}^C$	82.2
$q_{5,5,2}$	0.0	$t_{3,5,1}^H$	65.6	$t_{2,5,1}^C$	157.23125572159418
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,2}^C$	148.34264863679172
$q_{cu,2}$	0.0	$t_{4,1,2}^H$	271.1	$t_{3,1,1}^C$	160.0
$q_{cu,4}$	0.0	$t_{4,2,1}^H$	204.06666666666666	$t_{3,1,2}^C$	90.20394736842071
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,2,1}^C$	155.89423838982862
$q_{hu,2}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	221.1
$q_{hu,4}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	82.2
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	271.1	$t_{3,4,2}^C$	82.2
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,5,1}^C$	204.4
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	126.72824858757059	$t_{4,1,1}^C$	60.0
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,2,1}^C$	184.8311475409836
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,3,1}^C$	37.8
$t_{1,5,1}^H$	160.0	$t_{5,4,2}^H$	65.6	$t_{4,4,1}^C$	82.2
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{5,5,2}^H$	65.6	$t_{4,5,1}^C$	204.4
$t_{2,2,2}^H$	187.8424528301887	$t_{1,1,1}^C$	160.0	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	242.8630900490452	$t_{1,2,1}^C$	155.89423838982862	$t_{5,1,1}^C$	160.0
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	90.20394736842104
$t_{2,4,1}^H$	248.9	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	155.89423838982862
$t_{2,5,1}^H$	248.9	$t_{1,3,2}^C$	215.35238095238094	$t_{5,2,2}^C$	115.6
$t_{2,5,2}^H$	198.85754716980975	$t_{1,4,1}^C$	114.88574289844958	$t_{5,3,1}^C$	221.1
$t_{3,1,1}^H$	123.99729729729725	$t_{1,4,2}^C$	82.2	$t_{5,3,2}^C$	37.8
$t_{3,1,2}^H$	65.6	$t_{1,5,1}^C$	204.4	$t_{5,4,2}^C$	82.2
$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	204.4

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	124.01079136690646	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	2243.0899999999947	$b_{4,1,1}^{H,out}$	3415.8600000000006
$b_{1,3,1}^{H,in}$	1408.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	1614.4323555888961	$b_{4,2,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	476.493669902888	$b_{5,5,2}^{H,in}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	2243.0899999999947
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	1408.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	800.714212421227	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	1614.4323555888964
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	3355.16	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	1182.034345127225	$b_{2,1,2}^{H,out}$	476.493669902888	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	800.714212421227	$b_{1,3,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.8600000000006	$b_{2,5,2}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	2841.1600000000003
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	3355.16	$b_{1,5,2}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	1182.034345127237	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	0.0
$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0	$b_{2,3,1}^{C,in}$	1857.24

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	2841.1600000000003	$b_{5,4,2}^{C,out}$	1111.1700317460313
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	705.16	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	0.0		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	1857.24		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	103.82399999999976	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	705.15999999999879		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	103.82399999999976		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0		
$b_{5,4,2}^{C,in}$	1111.1700317460313	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		
$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0		